

Nondeterministic Finite Automata

Definition: nondeterministic finite automaton

A *nondeterministic finite automaton* (NFA) is a quintuple $N = (K, \Sigma, \Delta, s, F)$ where:

- K is a finite set of states,
- Σ is an alphabet,
- $s \in K$ is the initial state,
- $F \subseteq K$ is the set of final states, and
- $\Delta \subseteq K \times (\Sigma \cup \{e\}) \times K$ is called the transition relation.

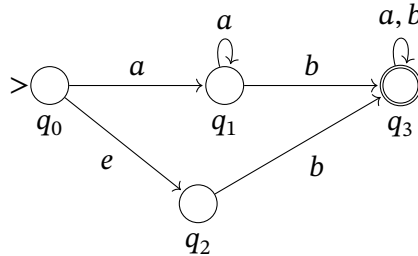
Definition: configuration and computation (NFA)

Let $N = (K, \Sigma, \Delta, s, F)$ be an NFA. A *configuration* of N is an element of $K \times \Sigma^*$. We say that a configuration (q, w) *yields* (q', w') in one step, written $(q, w) \vdash_N (q', w')$, if either

- $w = aw'$ for some $a \in \Sigma$ and $(q, a, q') \in \Delta$; or
- $w = w'$ and $(q, e, q') \in \Delta$.

Let $\vdash_N^*$ be the reflexive, transitive closure of $\vdash_N$. A string w is *accepted* by N if there exists some $q \in F$ with $(s, w) \vdash_N^* (q, e)$. The language accepted by N , $L(N)$, is the set of all strings accepted by N .

Q1. Let N be the NFA below over the alphabet $\{a, b\}$.



- (a) Write out the full sequence of configurations attained during the operation $(q_0, ab) \vdash_N^* (q_3, e)$.
- (b) Write out a different sequence of configurations starting at (q_0, ab) that does not end in an accepting configuration.
- (c) Is $ab \in L(N)$? Justify your answer using parts (a) and (b).

Q2. Write down the full specification of the NFA from Q1 from the definition.

Q3. Let $\Sigma = \{a, b\}$ and $k \geq 1$. Define

$$L_k = \{w \in \Sigma^* : |w| \geq k \text{ and the } k\text{th symbol from the end of } w \text{ is } a\}.$$

- (a) Design an NFA with $k + 1$ states accepting L_k . Explain, in a sentence, how nondeterminism is being used to recognize the language here.
- (b) Why would you expect a DFA for L_k to need far more than $k + 1$ states? Give a short reason for your answer.

Q4. Let $\Sigma = \{a, b\}$, and let L be the language of all strings that can be written as $w_1 w_2$, where w_1 has an odd number of a 's and w_2 has an even number of b 's.

- (a) Construct an NFA accepting L .
- (b) Prove that your NFA is correct, i.e. that $L(N) = L$.

Definition: e -closure

Let $N = (K, \Sigma, \Delta, s, F)$ be an NFA. For $q \in K$, define

$$E(q) = \{p \in K : (q, e) \vdash_N^* (p, e)\},$$

the set of states reachable from q without reading any input.

Q5.

- (a) Compute $E(q_0)$ for the NFA in Q1.
- (b) State and prove a necessary and sufficient condition on N , in terms of $E(s)$, for $e \in L(N)$ to hold.

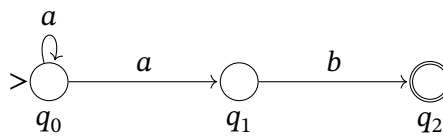
Theorem (Equivalence of NFAs and DFAs)

For each nondeterministic finite automaton, there is an equivalent deterministic finite automaton.

Let $N = (K, \Sigma, \Delta, s, F)$ be an NFA. An equivalent DFA $N' = (K', \Sigma, \delta', s', F')$ is given by

$$\begin{aligned} K' &= 2^K, \\ s' &= E(s), \\ F' &= \{Q \subseteq K : Q \cap F \neq \emptyset\}, \\ \delta'(Q, a) &= \bigcup_{q \in Q} \{E(p) : (q, a, p) \in \Delta\} \end{aligned}$$

Q6. Let N be the NFA below over the alphabet $\{a, b\}$.



- (a) Construct a DFA that is equivalent to N , using the construction above.
- (b) One of the states of your DFA is $\emptyset$. Explain why this state must be kept, unlike a state that is simply unreachable.
- (c) Which states of your DFA are final? In general, for a DFA obtained via this construction, can $\emptyset$ ever be a final state?

Q7. Construct a DFA that is equivalent to the NFA from Q1, using your answer to Q5(a) for $E(q_0)$.

- (a) Give your constructed DFA.
- (b) Unlike in Q6, the starting state of your DFA here already contains more than one state of N , before any input has been read. Explain why this happens for this NFA but not for the one in Q6.

Q8. An NFA N is called *unambiguous* if every input string has at most one accepting computation.

- (a) Give an example of an NFA over $\{a, b\}$ that is ambiguous.
- (b) Give an unambiguous NFA accepting the same language as your example in (a).
- (c) Do you think every language accepted by some NFA is also accepted by some unambiguous NFA? Give a short reason for your answer.

Q9. In Tutorial 2, we established that the set of all DFAs over a fixed alphabet Σ is countably infinite. Argue informally that the set of all NFAs over Σ is countably infinite as well.